

Do AI models of DNA actually understand biology?

MINA: linear evaluation reveals coding-sequence family signal in frozen DNA encoders up to protein composition

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Paper & code

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The question

DNA “foundation models” are increasingly used as off-the-shelf feature extractors, and often treated as interchangeable. **MINA asks a simple question:** what do their frozen embeddings *already* know, with no extra training — real biology, or surface patterns?

How we tested it

3,244 human genes in 5 families

● TF ● GPCR ● Kinase
● Ion channel ● Immune receptor

4 frozen DNA encoders

DNABERT-2 · NT-v2 ·
GENA-LM · HyenaDNA
weights never trained →
a fixed list of numbers

Simple linear probe
the only part that learns

Two read-outs

Q1 · which protein family?
Q2 · predict the gene’s text description

Benchmarked against **counting amino acids** (composition) and
a **real protein model** (ESM-2)

Words to know

Encoder — an AI model that turns a DNA sequence into a list of numbers.

Frozen — the model is used as-is; we never retrain it.

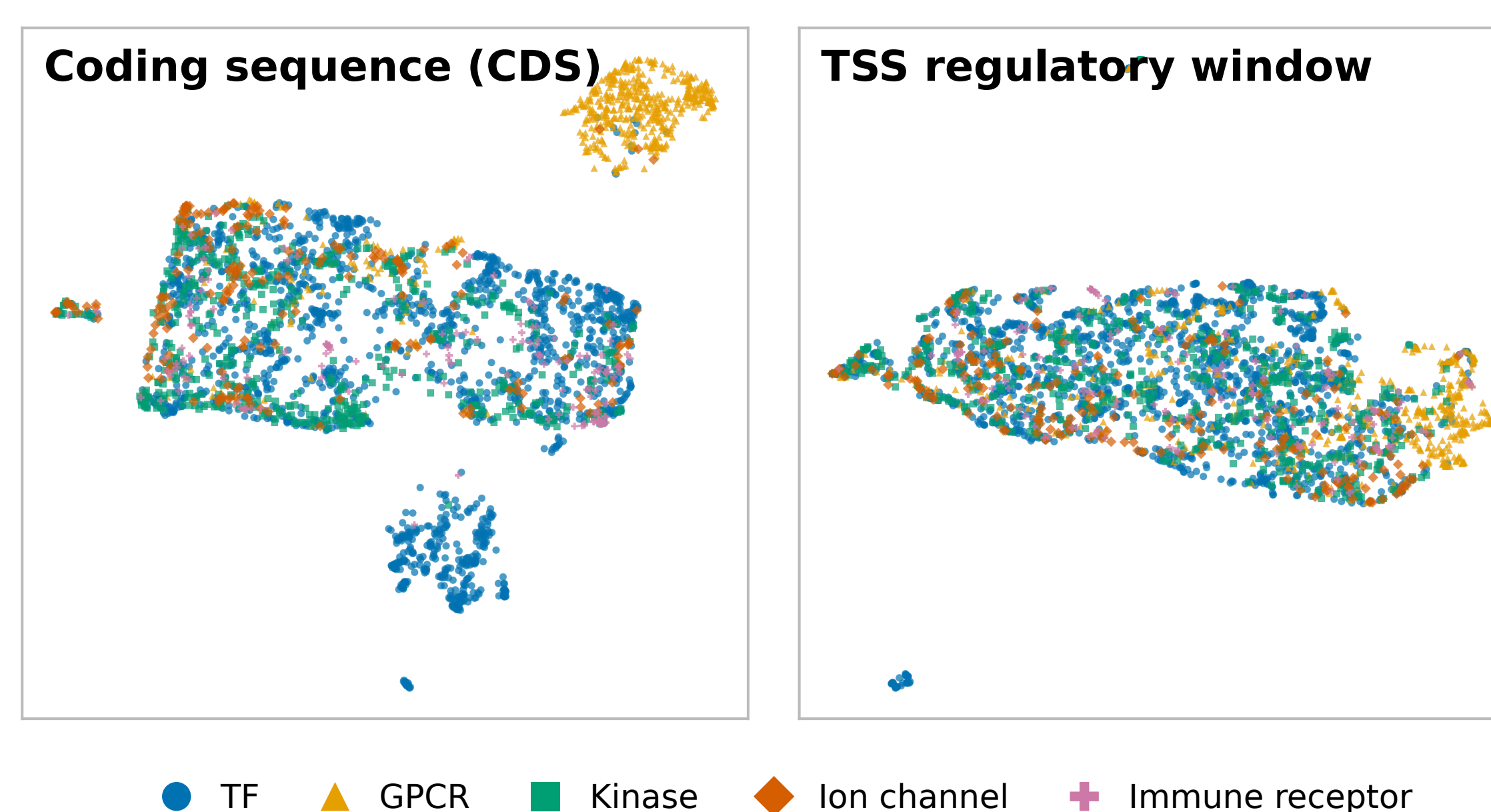
Linear probe — a tiny read-out we do train, to test what the numbers already contain.

CDS — a gene’s protein-coding sequence. **TSS window** — the region around a gene’s start, mostly regulatory (non-coding).

Composition — how often each short building block (k-mer or amino acid) appears, ignoring order.

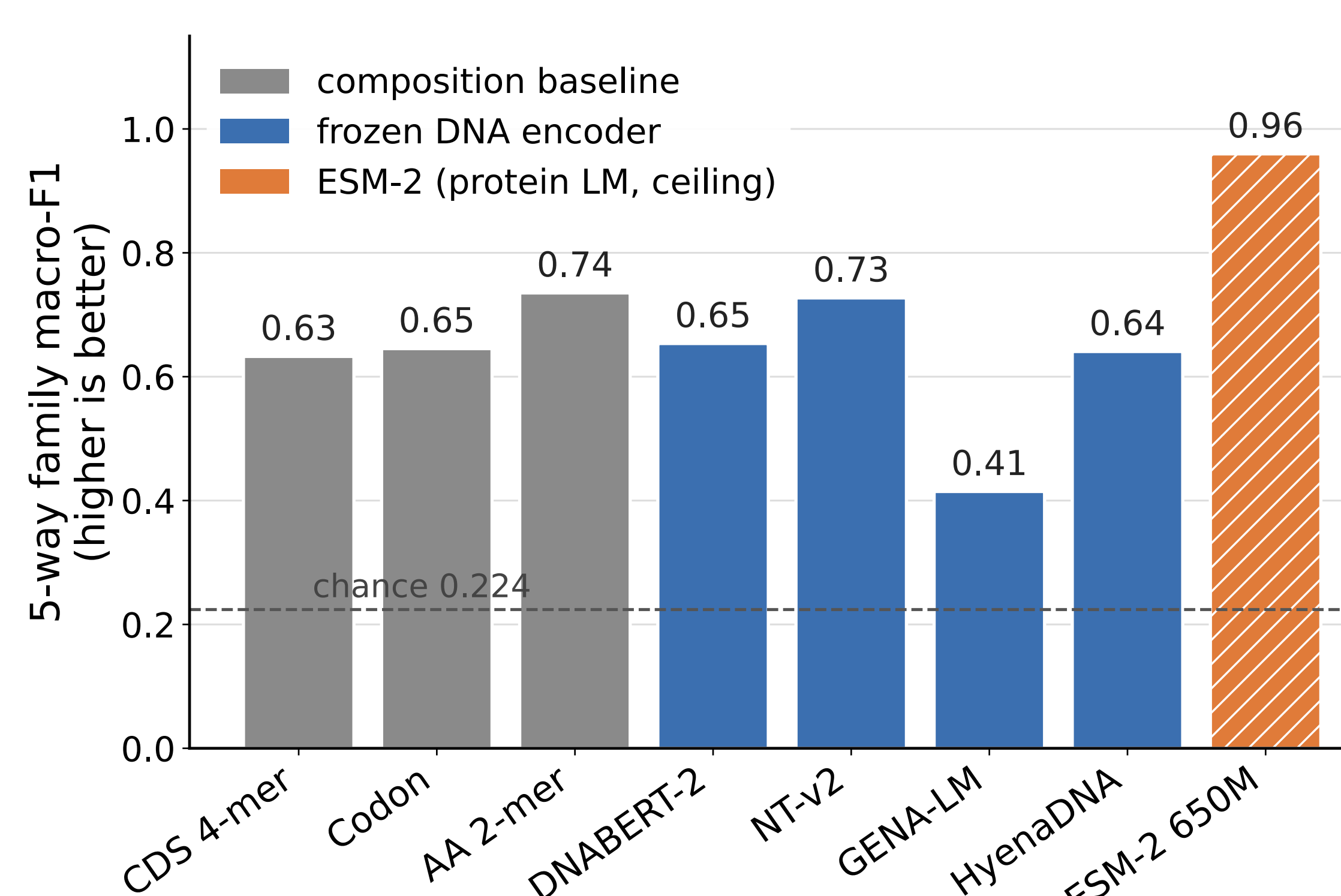
Homology-aware split — keep near-identical genes on the same side of train/test, so the model can’t just recognise a look-alike.

Coding genes carry a family signal



NT-v2 embeddings. From a gene’s **coding sequence**, GPCRs (orange) split off cleanly and family structure is visible; feed the same genes’ **regulatory window** and the families dissolve into one cloud.

... but only up to composition

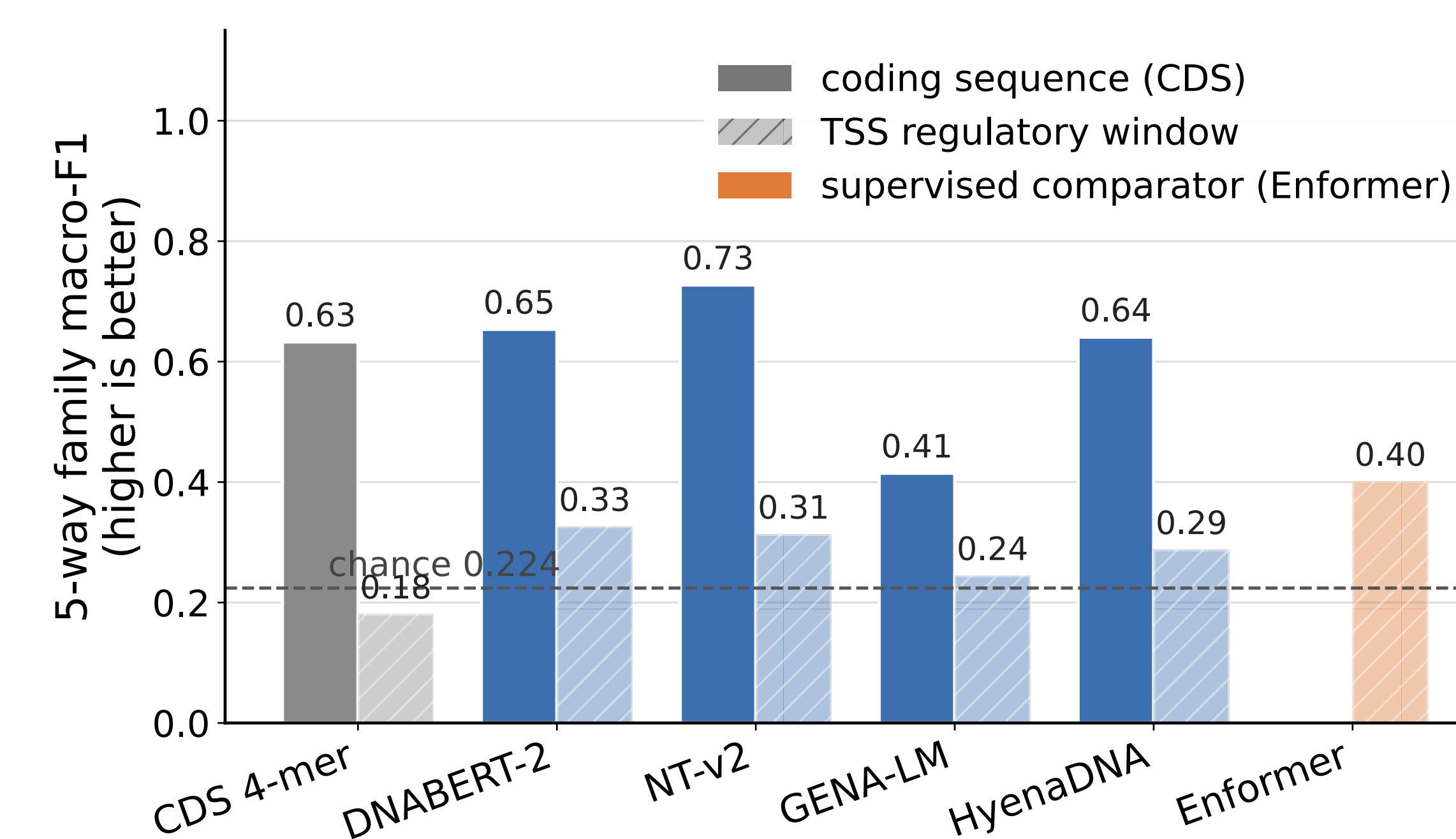


Grey = simple counting baselines · blue = the DNA AI models tested · orange = a specialized protein model (gold standard).

The best AI model (0.73) only ties plain amino-acid counting (0.74), and trails the protein model (0.96). GENA-LM (0.41) even falls below the baseline — the encoders aren’t interchangeable.

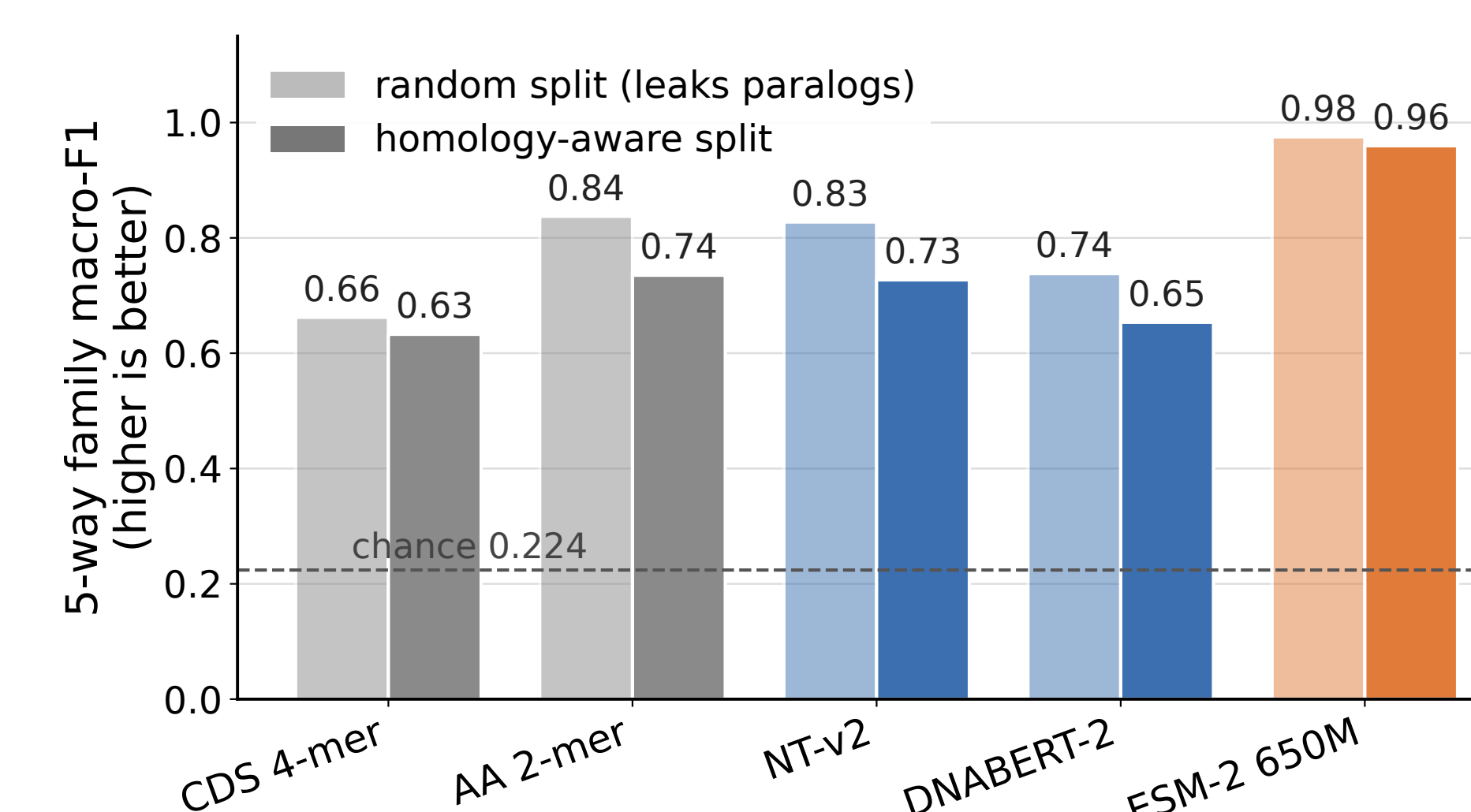
Paired test: vs. 4-mer +0.094 [+0.030, +0.166]; vs. amino-acid composition −0.008 [−0.064, +0.056], a statistical tie.

Regulatory context collapses



Swap coding sequence for the gene’s 196,608 bp regulatory (TSS) window and family signal falls to near chance (NT-v2 **0.313** vs. 0.224). Predicting a gene’s natural-language description is weak everywhere (best $R^2 = 0.077$ on coding sequence; near zero on the TSS window). The same collapse holds under a stricter *disjoint*-TSS split (NT-v2 0.313 → 0.259).

Honest evaluation matters



A naive **random** split inflates scores by letting near-identical genes leak between train and test. ESM-2 barely moves (0.96); the DNA encoders drop most (NT-v2 0.83 → 0.73), because part of their apparent skill was leaked look-alikes.

So what? Don’t treat DNA encoders as generic gene-function models. Trust a benchmark only if it has **(1)** a translated-composition baseline and **(2)** a homology-aware split — otherwise memorised similarity masquerades as understanding. **MINA is a cheap diagnostic that tells the two apart.**

3,244 human genes, 5 families · 4 frozen DNA encoders × 6 poolings (best pool) · homology-aware split (MMseqs2, 40% id) · macro-F1 vs. a shuffled-label chance floor (0.224), with 95% bootstrap CIs.

References: GenePT; ESM-2 (Lin et al.); Nucleotide Transformer; DNABERT-2.

Code: github.com/Austin-Senna/dna_to_text